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Health Messaging Strategies for Vaping Prevention and Cessation Among Youth and Young Adults: A Systematic Review

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ABSTRACT

This systematic review evaluates health messaging strategies for the prevention and cessation of e-cigarette use among youth and young adults. Health messaging strategies were defined as the strategic process of developing messages with the intent to shape, reinforce, or change recipients' health attitudes and behaviors. McGuire's Communication/Persuasion Model guided the analysis of the messaging strategies, focusing on the model's five communication inputs (i.e. source, message, channel, audience, destination) and 14 persuasive outcomes. Nine databases were searched from January 2007 to September 2023. The inclusion criteria encompassed studies in English that presented quantitative data on messaging strategies aimed at discouraging vaping among youth and young adults. Each study was also coded for study characteristics and the utilization of theory. Out of 6,045 studies, 25 met the inclusion criteria. The reviewed studies exhibit a diverse array of research methods and a consistent integration of theories. The review emphasizes the nuanced main and interaction effects of various communication inputs, such as message features and audience characteristics, while also pointing out a research gap in message sources. In addition, the utilization of social media for effective messaging to engage the audience requires further research. Only one study specifically evaluated messaging strategies for vaping cessation. More research is imperative to develop targeted and tailored messages that effectively prevent and reduce vaping, especially among populations at higher risk of vaping-related harms, while also leveraging effective channels and innovative communication technologies to engage the audience.

E-cigarette use is becoming increasingly prevalent and is emerging as a significant public health issue worldwide (Tehrani et al., 2022). In the United States, for example, the rapidly increasing use of e-cigarettes among youth has been referred to as a national vaping epidemic (Farzal et al., 2019). In 2023, over 2 million U.S. middle and high school students reported current (past 30-day) e-cigarette use, with 25% of current users vaping daily (Birdsey et al., 2023). Worldwide, youth e-cigarette use has risen significantly over the past decade. A study that analyzed data from 47 countries between 2015–2018 found that approximately 1 in 12, or 8.6% of youth reported vaping in the past 30 days (Chan et al., 2022). Compared to older adults, young adults aged under 25 were also more likely to be current users of e-cigarettes (Kramarow & Elgaddal, 2023).

The health risks associated with e-cigarettes are numerous and potentially severe. E-cigarettes produce toxicants, including acetaldehyde, acrolein, and formaldehyde, which contribute to pulmonary and cardiovascular disease development (Jin et al., 2021). People who use e-cigarettes are at higher risk for developing respiratory conditions, such as asthma (Xie et al., 2020). Furthermore, most e-cigarettes contain

nicotine, a highly addictive substance that can harm the developing brain of youth, which continues until the early to mid-20s (England et al., 2015). Moreover, e-cigarette use is associated with the initiation of combustible cigarette smoking among youth (Hammond et al., 2017; O'Brien et al., 2021; Soneji et al., 2017), raising concerns due to the well-established detrimental health effects associated with cigarette smoking (U.S. Department of Health and Human Services, 2014).

A wealth of evidence supports the role of health communication in reducing tobacco use (Allen et al., 2015; Murphy-Hoefer et al., 2020). Nevertheless, there are instances where anti-tobacco communication may inadvertently contribute to the stigmatization of the target audience, result in negative psychological experiences, and impact message effectiveness (Clayton et al., 2019; Kim, 2019; Riley et al., 2017). Consequently, it is imperative to develop anti-vaping messaging strategies that maximize message effectiveness while minimizing unintended consequences. We adopted the definition of health messaging strategies as the strategic process of developing messages with the intent to shape, reinforce, or change recipients' health attitudes and behavior (Kidd et al.,

2019). Moreover, the analysis of the messaging strategies in this review was guided by McGuire's Communication/Persuasion Model (2013), which encompasses five communication inputs including source, message, audience, channel, and destination (the desired behavioral change outcome). To the best of our knowledge, no systematic review has yet comprehensively examined messaging strategies in health communication aimed at vaping prevention and cessation among youth and young adults. Our review seeks to fill this research gap.

Literature review

McGuire's Communication/Persuasion Model

McGuire's Communication/Persuasion Model (2013) has influenced the design and evaluation of health communication promoting healthy behaviors (Arnold et al., 2022). The model is based on communication inputs and outputs, in which the inputs include components that go into designing a health message. Communication outputs refer to the series of cognitive, attitudinal, and behavioral changes that are triggered by a health message (McGuire, 1984, 2013). Table 1 provides an overview of McGuire's Communication/Persuasion Model, outlining key variables and descriptions. Specifically, the communication inputs include the source, message, channel, audience, and destination. The outputs illustrate the stages of audience response and engagement, ranging from exposure to the message, attention, liking or interest, comprehension, cognitive elaboration, skills acquisition, attitude change, memorization, recall/recognition, decision-making, behavioral change, maintenance, and advising others to behave likewise, comprehension,

cognitive elaboration, and skills acquisition, to more advanced levels such as attitude change, memorization, recall/recognition, decision-making, behavioral change, maintenance of the integrated action into long-term behavior, as well as advising of others to behave similarly.

Theoretical framework for health messaging research

Although the Communication/Persuasion Model explicates the components of a health message and its attitudinal and behavioral outcomes, the model does not elucidate the theoretical basis for the message effect. In this systematic review, we rely on Cappella (2006) which categorized theories applied to health behavior prevention and promotion into three classes: behavior change, message effect, and human information processing theories (Table 2). Behavior change theories provide insight into the pathways leading to behavioral intention and change but do not prescribe the specific design of messages required to exert changes. Message effects theories, on the other hand, focus on message format and content but may not be generalizable across different groups. Information processing theories provide insights into how audience characteristics interact with message features but do not fully account for social factors in the communication process. Cappella (2006) suggested that by integrating these three classes of theories, researchers can gain a more comprehensive understanding of how health messages are processed, how they influence behavior change, and how different audience characteristics can affect persuasion.

Table 1. The McGuire's Communication/Persuasion Model outputs.

Variable	Description
Inputs	
Source	The individual, group, or organization that the audience perceives as the communication originator
Message	What is being communicated and how it is being conveyed
Channel	The medium or platform through which the message is being transmitted
Audience	The intended target group, including their demographics, attitudes, and current behaviors, among other relevant factors
Destination	The desired outcome of the communication, which may involve changes in the audience's attitudes, beliefs, or behaviors in a specific domain
Outputs	
Exposure	Exposure to the message
Attention	Paying attention to the message
Liking or interest	Liking and being interested in the message
Comprehending	Understanding the message
Cognitive elaboration	Generating related cognitions to the message
Skills acquisition	Gaining the appropriate skills to act on the message
Attitude change	Agreeing with the communication's position
Memorization	Saving the message to memory
Recall/recognition	Retrieval of the message from memory when needed
Decision-making	Acting on the message
Behavioral change	Performing the action
Maintenance	Integration of the action into long-term behavior
Advising others to behave likewise	Proselytizing other people to behave likewise

Table 2. Three classes of theories for health messaging, adapted from Cappella (2006).

Theory class	Definitions	Examples
Behavior change theory	Theories describing rational, emotional, social, and personal predictors of healthy and risky behavior	The theory of planned behavior (Ajzen, 1991); Transtheoretical Model (Prochaska & Velicer, 1997)
Message effect theory	Theories predicting the format and content of messages that produce effects on cognitive, attitudinal, and emotional outcomes.	Emotional appeals (Nabi, 2007); Message frames (Rothman & Salovey, 1997)
Human information processing theories	Theories pertinent to the process of behavior changes	Limited capacity model of motivated mediated message processing (Lang, 2000); Elaboration likelihood model (Petty & Cacioppo, 1986)

Following McGuire's Communication/Persuasion Model (McGuire, 2013) and Cappella's theoretical framework (Cappella, 2006), this review systematically explores the literature to gain a comprehensive understanding of how communication inputs, both independently and interactively, influence each step of communication outputs in the context of anti-vaping communication aimed at youth and young adults. Moreover, we investigated the utilization of three classes of theories (i.e., behavioral change, message effect, and human information processing theories) in anti-vaping health messaging research for youth and young adults.

Method

Literature search

Search strategies and terms were devised in consultation with a librarian at a major university to include published research studies that focus on vaping-related health messages among youth and young adults. The search spanned from January 2007, coinciding with the introduction of e-cigarettes in the U.S. market (Sapru et al., 2020), to September 2023.

The updated PRISMA guidelines for reporting systematic reviews were used to guide the current study (Page et al., 2021). We conducted a comprehensive search in nine databases that were commonly used in previous systematic review studies including 1) Communication & Mass Media Complete, 2) Psychology & Behavioral Sciences Collection, 3) PubMed, 4) Embase, 5) Cumulated Index to Nursing and Allied Health Literature, 6) APA PsycInfo, 7) Education Resources Information Center, 8) Web of Science, and 9) Applied Social Sciences Index and Abstracts. Adapting a previous meta-analysis on the effects of vaping prevention messages on youth and young adults (Ma et al., 2023), we employed the following key terms related to e-cigarettes and health messaging aimed at youth and young adults: (adolescen* OR teen* OR "young adult*" OR youth*) AND (media OR mes-sag* OR campaign* OR advertis* OR educat* OR Prevent* OR "text messaging" OR "texting" OR communicat* OR PSA OR "public service announcement" OR warn* OR "counter mar-keting" OR counter-marketing OR counter advertis* OR coun-ter-advertis*) AND (vap* OR "electronic nicotine delivery system*" OR e-cig* OR ecig* OR "juul").

Inclusion criteria

All articles that were considered for inclusion had to meet the following criteria to be included in the review: (1) the study focused on one or more messaging strategies (i.e., message source, content, channel, or audience characteristics) within anti-vaping (i.e., destination) communication. Exclusions encompassed studies that merely compared exposure to an anti-vaping campaign/intervention with a control condition that did not examine any messaging strategies (Kowitt et al., 2023; Williams et al., 2022). This decision was made because a previous meta-analysis has already explored the impact of exposure to vaping prevention mes-sages compared to control among youth and young adults (Ma et al., 2023). Consequently, our review concentrates on

comparing diverse approaches within messaging strategies, such as the effects of different message sources employed in vaping prevention/cessation messages. (2) the study targeted youth and/or young adults for the anti-vaping messages. Therefore, research centered on messaging strategies for teachers, and parents was excluded from this review (e.g., Lazaro et al., 2021); (3) additionally, research on health warnings (e.g., Andrews et al., 2019) was excluded, as these messages often pertain to e-cigarette promotions, falling outside the scope of the current study and potentially merit-ing a separate systematic review. An exception was made for Rohde et al. (2022), as the authors emphasized that their study assessed the relative efficacy of warning themes, not specific product warnings intended for implementation on packaging and advertising; (4) the study had to report ori-ginal data collected from youth or young adults using quan-titative methods, excluding content analysis research, commentary or review (e.g., Kresovich et al., 2022; Xuan & Choi, 2021) and study protocol with no data reported (Lyu et al., 2022); (5) lastly, the study has to be quantitative research, excluding qualitative studies (e.g., Cavallo et al., 2019). The preference for quantitative research is based on the aim of quantifying and assessing the impact of different messaging strategies in anti-vaping health communication.

The initial database search yielded 6,045 studies. After removing duplicates, 2,255 studies remained. The first author screened article titles and abstracts to determine potential relevance. The full texts of the remaining 146 articles were located and reviewed by the first author. Reasons for study exclusion were tracked. Overall, a final set of 25 peer-reviewed journal articles met inclusion criteria and were included in the systematic review. Figure 1 displays the PRISMA flow diagram that depicts the retrieval and inclusion/exclusion process.

Article coding

Adapted from a previous systematic review of public health communication for a range of non-cigarette tobacco products (Ross et al., 2019), articles were first coded by study character-istics including research method, study sample, and study design. Following McGuire's Communication/Persuasion model (McGuire, 2013), studies were further coded by com-munication inputs and outputs. Articles were additionally coded for the three types of theories including behavior change, message effect, and human information processing theories (Cappella, 2006). Supplemental Table S1 lists the operational definitions for each of the coding categories.

Study characteristics

Studies were analyzed based on 1) the research method used (i.e., experimental study, survey); 2) the study sample, mainly the participant age range, country of study, the sample size, and if the study focused on a specific population group among youth and young adults, and 3) the study design (i.e., cross-sectional, longitudinal).

Theories

We coded for the use of theories that are central to health messages (Cappella, 2006). Specifically, whether the study

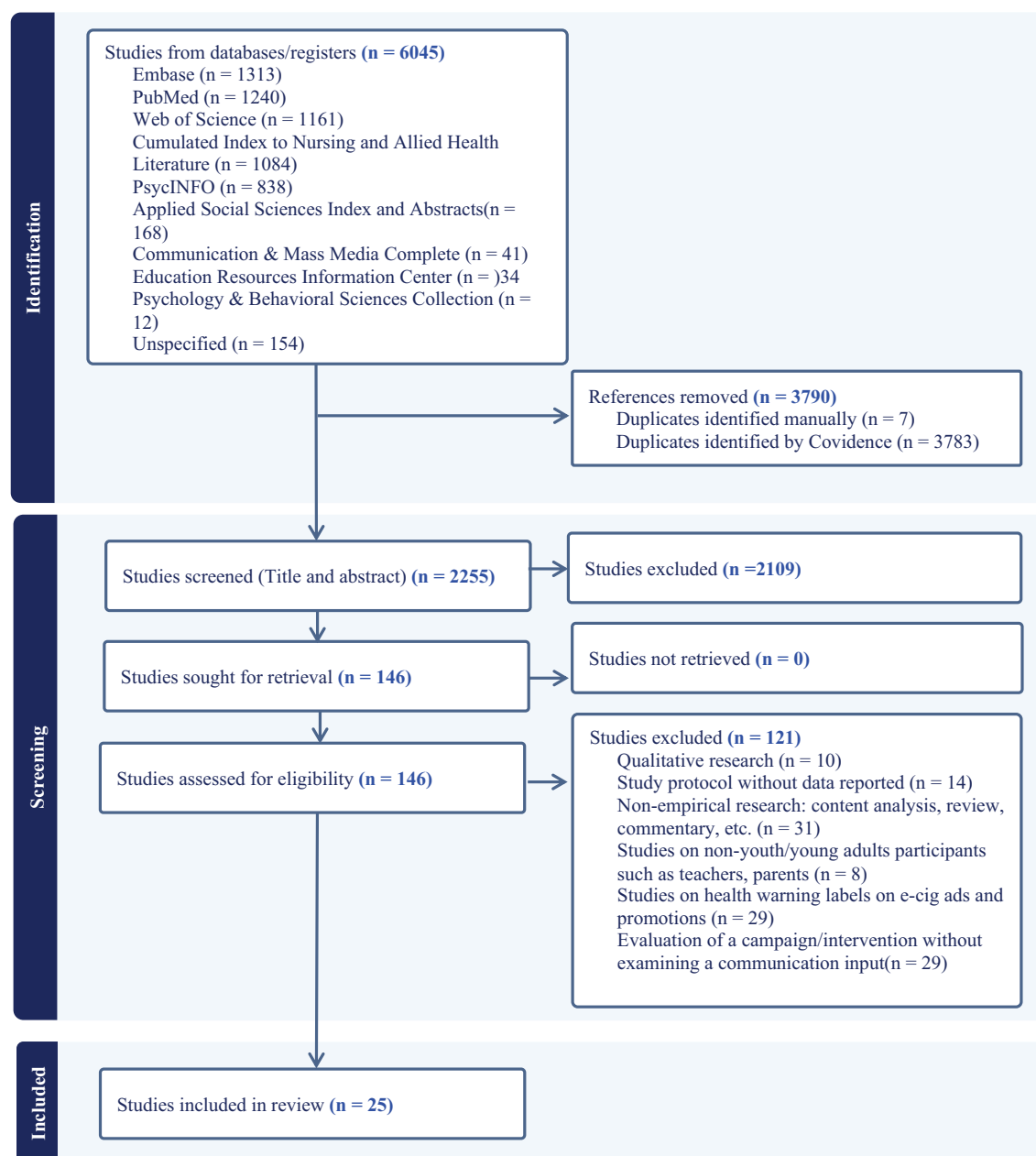


Figure 1. PRISMA flow diagram showing the study screening process.

included any of the three types of theories: behavior change, message effect, and human information processing theories.

Main, stratified, and interaction effects of communication inputs on outputs

Given the different approaches of data analyses, studies were coded to examine the main, stratified, and interaction effects of five communication inputs on communication outputs (McGuire, 2013). The communication inputs include 1) source, 2) message, 3) channel of communication, 4) audience, and 5) destination, representing the desired outcome of communication. Each study was coded based on whether it centered on vaping prevention or cessation. For a cessation destination, a study had to explicitly state its message aimed at facilitating quitting vaping among youth and/or young adults, exclusively

enrolling participants currently using e-cigarettes. Other studies were categorized under a vaping prevention destination.

In line with McGuire's Communication/Persuasion Model (2013), communication outputs were assessed across 12 persuasive outcomes, including attention, liking or interest, comprehending, cognitive elaboration, skills acquisition, attitude change, memorization, recall/recognition, decision-making, behavioral change, maintenance, and advising others to behave similarly. The exposure communication output variable was excluded, as it typically serves as the starting point in the information process and has received less attention as an outcome of message effects. Additionally, two frequently measured persuasive outcomes, perceived message effectiveness (PME, Noar et al., 2021), and positive and negative affective response (Nabi, 2007), were also coded in the studies.

Results

Study characteristics

A summary of the study characteristics and theoretical framework is presented in Table 3. Among the 25 studies examined, five utilized a longitudinal research design to evaluate the effects of anti-vaping messages over varying periods. These durations included six days (Noar et al., 2019), one week (Xu et al., 2023), three weeks (Noar et al., 2022), 30 days (Calabro et al., 2019), and 7 months (Graham et al., 2020). Only one study (Graham et al., 2020) focused on vaping cessation by recruiting only current e-cigarette users who had an intention to quit in the past 30 days.

The participants in the 25 studies ranged in age from 13 to 30. Sample sizes varied from 69 to 5,405. All studies were conducted in the U.S. Four of these studies specifically targeted certain population groups rather than the general youth or young adult population. Specifically, Cartujano-Barrera et al. (2022) focused on Black and Latino adolescents, Lee et al. (2022) targeted sexual minority young adults, Patterson et al. (2022) concentrated on sexual minority women and nonbinary individuals assigned female at birth, and Tan et al. (2018) examined current smokers and dual users among young adults.

Theoretical framework

In this review, 17 out of 25 studies incorporated a theory. *Message effect theories* were examined in 10 studies. Specifically, drawing from the prospect theory, 6 studies compared gain-framed and loss-framed anti-vaping messages, analyzing audience reactions and perceptions of the messages (Cai & Zhao, 2021; Calabro et al., 2019; Jun et al., 2021; Kong et al., 2016; S. Liu & Yang, 2020a; Zhao et al., 2023). Three studies analyzed the effects of narrative in anti-vaping communication through the message narrative theory (S. Liu & Yang, 2020a, 2020b; Zhao et al., 2023). One study utilized the source credibility theory to examine how peer and expert message sources affect message effectiveness (Lee & Stevens, 2022).

Six studies applied *human information processing theories*. The Limited Capacity Model of Motivated Mediated Message Processing (LC4MP) was tested in two studies to examine how message features affect cognitive elaboration (Clayton et al., 2020; Sanders-Jackson et al., 2019). One study examined the psychological reactance theory in the process of anti-vaping communication among the audience (Clayton et al., 2020). The theory of psychological distance was also utilized in two studies to investigate how narratives and different VR content influence message perceptions among the audience (S. Liu & Yang, 2020b; Xu et al., 2023). The Associative Learning Theory was applied in one study to test how vapors in anti-vaping PSAs may elicit vaping and smoking urges (Sanders-Jackson et al., 2019).

Three studies examined the effects of audience characteristics based on *behavior change theories*, including one study that tested individuals' uncertainty tolerance (Cai & Zhao, 2021); one study examined the effects of self-affirmation and group-affirmation on college students' attitudes toward a vape-free policy (Jun et al., 2021). In addition, the social

identity theory was used to analyze how one's sexual orientation may affect and interact with message themes among sexual minority individuals (Lee et al., 2022).

While not empirically examining the theoretical framework or testing hypotheses, the remaining studies developed anti-vaping communication based on various theoretical frameworks or examined the effectiveness of national vaping prevention campaigns such as The Real Cost E-Cigarette Prevention Campaign (Noar et al., 2022). Those theories include the social cognitive theory (Graham et al., 2020), the biobehavioral model of nicotine addiction (Cartujano-Barrera et al., 2022), the "mix of attributes" theory-based design (Lazard, 2021), and the theory of reasoned action (Noar et al., 2022; Rath et al., 2021).

Importantly, three studies incorporated at least two types of theoretical frameworks including message effect theory, behavior change theory, and human information processing theory (Cai & Zhao, 2021; Jun et al., 2021; S. Liu & Yang, 2020b), allowing the three studies to examine the interactions between theory-driven variables across message content, audience characteristics, and the psychological processing of anti-vaping messages. For example, integrating both message narrative theory and reactance theory, S. Liu and Yang (2020b) provided nuanced insights into how narrative messages influence the audience through the mediating role of the perceived threat to freedom among the audience. The other two studies shed light on how personality constructs including uncertainty tolerance and affirmation interact with message framing to affect the persuasive outcomes of messages (Cai & Zhao, 2021; Jun et al., 2021).

Main, stratified, and interaction effects of communication inputs

Table 4 shows the main and interaction effects of the communication inputs under study. The table presents all significant findings from each study related to the investigated communication inputs. The table also provides stratified results based on audience characteristics and/or interaction effects between two or more communication inputs on the measured outputs.

Source

Only one study examined the effects of different message sources on young adults' reactions to anti-vaping messages. Lee and Stevens (2022) aimed to investigate the impact of message source credibility on e-cigarette harm perceptions among young adults. The results showed that messages from expert sources were associated with higher perceived source credibility compared to messages from peer sources. In addition, higher source credibility led to increased harm perceptions of e-cigarettes. However, message sources did not significantly affect e-cigarette use intentions. In addition, the author did not find an interaction effect between message source (peer vs. expert) and vaping status (current use vs. former use vs. never use) on message effectiveness.

Message

Framing. Framing represents a focal point in the examination of message characteristics within anti-vaping communication

Table 3. Study characteristics and theoretical framework.

	Study characteristics			Theory		Destination
	Research method	Participant age range; Country of study	Sample size	Study design	Employed theory	
Boynton et al., (2022)	Content analysis of existing anti-vaping ads	13 to 17; US	1,501	Cross-sectional	n/a	No
Cai & Zhao, (2021)	Survey 2 (frame) × 2 (uncertainty tolerance) factorial experimental design	13 to 17; US	536	Cross-sectional	Message effect theory • Prospect theory Behavior change theory • Uncertainty tolerance	No
Calabro et al., (2019)	Two-group (gain vs. loss frame) randomized pretest and posttest experimental design	18 to 24; US	95	Longitudinal : participants received 4 texts weekly for 30 days	Message effect theory • Prospect theory	No
Cartujano-Barrera et al., (2022)	Four-group (four themes) randomized pretest and posttest trial	13 to 17; US Black and Latino adolescents	362	Cross-sectional	Behavior change theory • Biobehavioral model of nicotine addiction	No
Clayton et al., (2020)	2 (condition: dogmatic versus suggestive language) × 3 (audio probe) × 4 (message repetition) factorial experimental design	18 to 28; US	155	Cross-sectional	Human information processing theories • Psychological reactance theory • Limited capacity model of motivated mediated message processing (LC4MP)	No
Graham et al., (2020)	Two-group (This is Quitting intervention vs. control) randomized clinical pretest and posttest trial	18 to 24; US	2,588	Longitudinal : participants received monthly assessments for 7 months	Human information processing theories • Social cognitive theory	Yes
Jun et al., (2021)	2 (gain vs. loss frame) × 2 (self vs. group affirmation) factorial experimental design	College students; US	219	Cross-sectional	Message effect theory • Prospect theory Behavior change theory • Self-affirmation theory	No
Kong et al., (2016)	Survey	Middle/high school or university students; US	5,405	Cross-sectional	Message effect theory • Prospect theory	No
Lazard, (2021)	3 (message formats) × 6 (theme) factorial experimental design	15 to 18; US	928	Cross-sectional	Message effect theory • "Mix of attributes" theory-based design	No
Lee & Stevens, (2022)	Two-group (expert vs. peer message source) randomized experiment	18 to 30; US	302	Cross-sectional	Message effect theory • Source credibility theory	No
Lee et al., (2022)	Three-group (3 themes) randomized experiment	18 to 30; US Sexual minority individuals	457	Cross-sectional	Behavior change theory • Social identity theory	No
S. Liu & Yang, (2020a)	2 (narrative vs nonnarrative) × 2 (gain vs loss frame) factorial design experiment	College students; US	441	Cross-sectional	Message effect theory • Prospect theory • Message narrative	No
S. Liu & Yang, (2020b)	Two-group (narrative vs. non-narrative) randomized experiment	College students; US	439	Cross-sectional	Message effect theory • Message narrative Human information processing theories • Theory of psychological distance	No
J. Liu et al., (2023)	Survey	18 to 30; US	459	Cross-sectional	n/a	No
Noar et al., (2019)	Pretest-posttest within subject experiment	14 to 18; US	69	Longitudinal : participants received e-cigarette harms message per day for six days	n/a	No
(Noar et al., (2022)	Three-group (2 themes + 1 control) randomized clinical trial	13 to 17; US	1,514	Longitudinal : participants saw 3 randomly ordered 30-second video advertisements online at each of 3 weekly study visits	Behavior change theory • The theory of reasoned action	No

(Continued)

Table 3. (Continued).

	Study characteristics				Theory	Destination
	Research method	Participant age range; Country of study	Sample size	Study design		
Patterson et al., (2022)	3 (theme) by 3 (image format) factorial design experiment with a control condition	18 to 30; US Sexual minority women and nonbinary people assigned female at birth	501	Cross-sectional	n/a	No
Rath et al., (2021)	Survey	15 to 24; US	1,564	Cross-sectional	Message effect theory • The Hornik and Woolf health communication framework	No
Rohde et al., (2022)	Five-group (themes) randomized experiment	13 to 17; US	623	Cross-sectional	n/a	No
Sanders-Jackson et al., (2019)	2 (presence vs. absence of vapor) x 4 (message repetition) between subject experiment	21 to 30; US	71	Cross-sectional	Human information processing theories • Associative Learning Theory • LC4MP	No
Stevens et al., (2021)	Participants randomly view and rate six of the 32 images	18 to 24; US	200	Cross-sectional	n/a	No
Tan et al., (2018)	Four-group (themes) randomized experiment	21 to 30; US Smokers and dual users	293	Cross-sectional	n/a	No
Villanti et al., (2021)	2 (content) x 3 (theme) factorial experimental design	18 to 24; US	200; 769;	Cross-sectional	n/a	No
Xu et al., (2023)	Three-group randomized experiment	College students; US	137	Longitudinal : one time exposure, collect data pretest, immediate after the test, and 1 week later	Human information processing theories • Theory of psychological distance	No
Zhao et al., (2023)	2 (framing) x 2 (format) factorial design experiment	13 to 17; US	1,084	Cross-sectional	Message effect theory • Prospect theory • Message narrative	No

Table 4. Main and interaction effects of communication inputs.

	Studied communication inputs	Measured communication outputs	Main effects of communication inputs	Stratified and/or Interaction effects of communication inputs
Boynnton et al., (2022)	Message elements: - Themes - Imagery - Other features Audience vaping status: - Current users - Susceptible non-users - Non-susceptible non-users	• Interest or liking: vaping appeal • PME	Message elements: • Interest or liking (vaping appeal): • Flavor theme and imagery associated with food increased vaping appeal across all groups. • PME: • Themes Associated with Increased PME: nicotine addiction, chemicals, negative health effects, health symptoms, comparison to cigarettes, non-harmless water vapor, death, unknown health effects, gateway effects, warning symbols, graphic images, combustible cigarettes • Themes Associated with decreased PME: industry targeting, flavors, environmental impact, faces (both adult and teen), food • No effect: vaping device imagery	Stratified by audience vaping status: • Interest or liking (vaping appeal) • Imagery associated with food had no impact for current users and a modest positive effect for non-users. • PME • Effective among non-users but not users: the word "addiction," themes on brain effects, and non-harmless water vapor • Effective among users but not non-users: theme on lung effects • Effective among users and susceptible but not non-susceptible: themes on death and nicotine chemical symbol
Cai & Zhao, (2021)	Message framing: - Gain frame - Loss frame Audience uncertainty tolerance: - High tolerance - Low tolerance	• Positive affect • Negative affect • Attitudes: attitudes and risk beliefs of vaping • Decision-making: intention to vape • PME	Framing: Loss frame was overall better received and more effective compared to gain frame on: • Negative affect • Attitudes: risk beliefs of vaping • Decision-making • PME Uncertainty tolerance: Lower in uncertainty tolerance produced more desired outcomes on: • PME (single and multi-items), • Negative affect • Positive affect • Attitudes: risk beliefs of vaping	Framing × uncertainty tolerance: Gain framing is more effective for those with low uncertainty tolerance, while loss framing is more effective for those with high uncertainty tolerance on • PME (single and multi-items) • Attitudes (negative beliefs and attitudes) • Decision-making (intention to vape)
Calabro et al., (2019)	Message framing: - Gain frame - Loss frame	• Attitudes: risk beliefs of vaping • Memorization: knowledge about e-cig harm	Framing: Gain frame produced more desired outcomes on: • Attitudes (risk beliefs of vaping)	n/a
Cartujano-Barrera et al., (2022)	Themes: - Health rewards; - Financial rewards; - Autonomy; - Social norms	• Liking and interest: satisfaction and willingness to share with friends • Decision-making: susceptibility to future vaping	Themes: • None of the themes reduced participants' susceptibility to vaping in the future	n/a
Clayton et al., (2020)	Language style: - Dogmatic - Suggestive	• Cognitive elaboration: threat to freedom; reactance; counterarguing; encoding • Negative affect: anger • Decision-making: intention to vape	Language style: Suggestive messages led to more desired outcomes than dogmatic messages on • Cognitive elaboration: perceived freedom threats; counterarguments; response times; encoding of content • Negative affect: anger, • Decision-making: intentions to vape	
Graham et al., (2020)	Message intervention: - This is quitting - Control Audience characteristics: - Demographics - Nicotine dependence	• Behavioral change: self-reported 30-day point prevalence abstinence at 7 months	Message intervention: • Behavioral change: participants who received the intervention were one-third more likely to quit vaping at the 7-month primary end point compared with control participants.	Intervention × audience characteristics: The study did not find any significant interaction effects between the intervention and any audience characteristics, including nicotine dependence
Jun et al., (2021)	Message framing: - Gain frame - Loss frame Audience affirmation: - Self-affirmation - Group-affirmation Vaping status: - Vapers - Non-vapers	• Attitudes: Vape-free policy support	Framing: • No significant framing effects Audience affirmation: • Attitudes: self-affirmation is associated with a higher level of vape-free policy support than group-affirmation Audience vaping status: • Attitudes: Non-vapers reported higher levels of policy support than vapers	Framing × audience affirmation • No interaction effects Framing × audience affirmation × vaping status Attitudes: for vapers, the level of policy support was the highest when they were self-affirmed and exposed to a gain-framed message. For non-vapers, the level of policy support was the highest when they were self-affirmed and exposed to a loss-framed message

(Continued)

Table 4. (Continued).

	Studied communication inputs	Measured communication outputs	Main effects of communication inputs	Stratified and/or Interaction effects of communication inputs
Kong et al., (2016)	Message framing: (1) Gain frame (2) Loss frame Themes (1) Financial cost (2) Health risks (3) Addiction (4) Social labeling as a smoker Audience characteristics (1) Audience school level (2) Audience sex (3) Audience vaping status (4) Lifetime users (5) Never users	• Liking or interest: preference of the message	Framing: • Liking or interest: 1) Loss frame was preferred over gain frame for themes related to health risks, addiction potential, and social labeling as a smoker; 2) Gain frame was preferred over loss frame for the theme related to financial cost Audience characteristics: • Liking or interest: 1) females preferred loss-framed messages for all themes relative to males; 2) lifetime e-cigarette users preferred loss-framed health risks and social labeling messages relative to never users; 3) high school students preferred gain-framed social labeling messages relative to college students	Among never e-cigarette users: • Liking or interest: 1) females prefer loss-framed messages across all four message themes than males; 2) preference for gain- vs. loss-framed messages did not differ by cigarette susceptibility or e-cigarette susceptibility status
Lazard, (2021)	Message format: (1) Visual based, (2) Quiz (3) Text only Themes: (1) Lung damage (2) Contains chemical (3) Mood effect (4) Nicotine effect (5) Can explode (6) Vape designs Audience vaping status (1) Current users (2) Ever users (3) Susceptible to future use	• PME • Positive affect • Negative affect • Cognitive elaboration • Memorization: knowledge about e-cig harms • Attitudes: risk beliefs of e-cigs • Advising others to behave likewise: sharing with others	Message format: • No main effects Themes: Lung damages produced more desired outcomes on: • PME: compared to mood change; nicotine effects; vape design • Cognitive elaboration: nicotine effects; can explode; vape design	Message format/theme × vaping status • No interaction effects: e-cigarette use status did not moderate the effects of message theme or format on message reactions.
Lee & Stevens, (2022)	Message sources (1) Expert (2) Peer Audience vaping status (1) Current users (2) Former users (3) Never users	• Liking or interests: credibility • Attitudes: risk perceptions of e-cigs • Decision-making: intention to use e-cigs	Message source: The expert message produced more desired outcomes than peer message on: • Liking or interests: credibility	Message source × vaping status • No interaction effects
Lee et al., (2022)	Themes (1) E-cig health harms (2) Pride (3) Wellness Audience LBQ identity	• PME	Audience SGM identity: • PME: affirmation of a few gender minority (SGM) identities was significantly positively associated with PME	Message theme × SGM identity • PME: Positive effects of identity affirmation on PME were more pronounced when participants viewed pride-themed messages compared to wellness or harms messages
S. Liu & Yang, (2020a)	Language style (1) Narrative (2) Non-narrative Message framing (1) Gain frame (2) Loss frame	• Attitudes: risk perception of e-cigs • Decision-making: intention to use e-cigs	n/a	Narrative style × message framing • Decision-making: 1) Gain-framed narrative, compared to non-narrative led to lower intention to use e-cigs 2) Loss-framed narrative decreased intention to vape through stronger sadness and higher risk perception.
S. Liu & Yang, (2020b)	Narrative language style (1) Narrative (2) Non-narrative Audience vaping status (1) Ever users (2) Never users	• Positive affect • Negative affect • Attitudes: risk perception of e-cigs; perceived susceptibility; perceived severity • Decision-making: intention to use e-cigs	Narrative language style: Narrative messages produced more desired outcomes than non-narrative messages on: • Transportation → affect → attitudes (serial mediation)	Narrative style × vaping status • Attitudes: Participants who have not used e-cigs before report more negative attitude after reading the narrative message than those who have used e-cigs

(Continued)

Table 4. (Continued).

	Studied communication inputs	Measured communication outputs	Main effects of communication inputs	Stratified and/or Interaction effects of communication inputs
J. Liu et al., (2023)	Audience vaping status: (1) Never use of e-cigarettes (2) Past 30-day e-cigarette use (3) Ever use of e-cigarettes Audience social media use	• Liking and interest: engage with the posts	Audience characteristics: • Liking and interest: 1) E-cig use status (ever vs. current) did not affect participants' intention to engage (e.g., like, comment on) with anti-vaping social media posts; 2) Total number of social media sites used was associated with intentions to "like" the posts and the overall engagement score; 3) Daily internet use was associated with intentions to "comment on" and "like" the posts.	n/a
Noar et al., (2019)	Themes: (1) Chemical (2) Brain (3) Nicotine	• PME • Memorization: e-cig knowledge • Recall/recognition: message recognition • Cognitive elaboration • Attitudes: perceived risk • Decision-making: vaping susceptibility; willingness to use e-cig • Advising others to behave likewise: sharing with others	Themes: Chemical and brain messages produced more desired outcomes than nicotine messages on: • PME • Fear 2) Nicotine messages produced higher values than chemical messages on: • Recall/recognition	n/a
Noar et al., (2022)	Themes: (1) Health harm (2) Addiction (3) Control	Primary • Decision-making: vaping susceptibility Secondary • Negative affect • Cognitive elaboration • Attitudes: vaping health harm risk perceptions; vaping addiction risk perceptions; vaping attitudes • Advising others to behave likewise: social interactions	Themes: The Real Cost groups did not differ from one another on susceptibility and secondary outcomes to vaping and all other message reactions at visit 4.	n/a
Patterson et al., (2022)	Themes: (1) Harms (2) Wellness (3) Pride Image format: (1) Individual (2) Couple (3) Cartoon formats Audience vaping status (1) Current users (2) Former users (3) Never users Audience sexual identity	• PME • Attitudes: changes in feelings toward e-cigs; perceived threat • Decision-making: discouragement from vaping	Themes: Harm messages produced more desired outcomes than wellness and pride themes on: • Decision-making: intention to vape Image format: Female-presenting couples scored higher than cartoon images on: • PME	Stratified effect across vaping status: Theme • Decision-making: Current e-cig users reported greater intentions to abstain from vaping when they viewed harms-themed messages compared to current e-cig users who were in the control condition, the pride condition, and the wellness condition. Image format • Decision-making: Current e-cig users who viewed an image of an individual woman reported higher readiness to avoid vaping and higher intentions to abstain than those in the control condition, the cartoon condition, and the couple condition. Sexual identity × message theme/image format Sexual identity did not modify the effect of message theme and image format on any outcomes

(Continued)

Table 4. (Continued).

	Studied communication inputs	Measured communication outputs	Main effects of communication inputs	Stratified and/or Interaction effects of communication inputs
Rath et al., (2021)	Theme: (1) Social acceptability of vaping (2) Anti-vape industry sentiment (3) Independence from vaping (4) Non-vaping identity (5) Perceived risk of harm Audience vaping status: (1) Current users (2) Ever users (3) Susceptible never-users (4) Non-susceptible never-users	• Attitudes: anti-vape attitudes and beliefs	n/a	Message theme × vape use status • Attitudes: target message theme scales varied based on vape use status: 1) non-susceptible never-users had highest mean scale scores followed by susceptible never-users, ever users, and current users followed in decreasing order
Rohde et al., (2022)	Themes (1) Chemical harms (2) Lung harms (3) COVID-19 harms (4) nicotine addiction (5) control (messages about vape litter) Audience demographics	Primary • PME Secondary • Negative affect • Attention • Advising others to behave likewise: social interaction	Themes: • PME: Adolescents rated chemical, lung, and COVID-19 harms higher on PME than nicotine addiction and control while nicotine addiction was rated higher than control • Negative affect: Chemical, lung, and COVID-19 harms elicited greater negative affect than nicotine addiction and control	Message theme × vape use status No interaction effects on • PME: The warning message theme had no interaction with age, gender, race, Hispanic ethnicity, household income, sexual attraction, or vaping status
Sanders-Jackson et al., (2019)	Message feature: (1) Presence of vapor (2) Absence of vapor	• Cognitive elaboration: Psychophysiological measures • Positive affect • Memorization: knowledge of vaping risks • Decision-making: urge to vape • PME • Liking or interest	Vapor: The presence of vapor activates the approach/appetitive motivational system, leading to greater • Positive affect: • Cognitive elaboration • Memorization • Decision-making: urge to vape	
Stevens et al., (2021)	Message features (1) People (2) Vapor (3) Device (4) Color (5) Similarity to warning label image Audience vaping status (1) Current users (2) Non-users		Message feature: • Liking or interests: 1) Images with vapor, devices, or e-liquid, were not well-liked than images without those features; 2) warning label images were not well-liked than those without a warning • PME: Images with vapor, devices, or e-liquid, were perceived as more effective than without the features; 2) warning label images were perceived as more effective than those without a warning	Stratified effect across vaping status: • PME: 1) Presence of vapor was seen as more effective for non-users and users; 2) Presence of e-liquids and device was more effective and likable for non-users; 3) The presence of color was seen as less effective for non-users only
Tan et al., (2018)	Themes: (1) Anti-vaping PSAs with vapor (2) Anti-vaping PSAs without vapor (3) Physical activity PSAs (4) Antismoking PSAs with smoking cues Audience vaping status; (1) Smokers (2) Dual users of cigarettes and e-cigarettes	• Decision-making: changes in vaping urge; intention to vape in the next hour; Intention to purchase vapes; Changes in smoking urge; Intention to smoke in the next hour; Intention to purchase cigarettes	n/a	Stratified effect across tobacco use status: Smokers Only: • Decision-making: 1) Exposure to anti-vaping PSAs with vapor (vs. control) linked to lower intention to vape and purchase e-cigs, lower intention to smoke and purchase cigarettes; 2) Exposure to anti-vaping PSAs with vapor (vs. PSAs without vapor and vs. anti-smoking PSAs with smoking cues) linked to lower intention to vape; 3) Exposure to anti-vaping PSAs without vapor (vs. control) associated with lower vaping urge and lower intention to purchase ENDS Dual Users: • Decision-making: Exposure to anti-vaping PSAs without vapor (vs. anti-smoking PSAs) associated with lower intention to purchase ENDS.

(Continued)

Table 4. (Continued).

	Studied communication inputs	Measured communication outputs	Main effects of communication inputs	Stratified and/or Interaction effects of communication inputs
Villanti et al., (2021)	Message content: (1) Addiction (2) Harm Themes: (1) Alone (2) Flavor (3) Social	• PME • Liking and interest: likability of the message	Main effect of content themes: • PME: 1) Harm content messages produced higher PME than addiction content messages; 2) Adding flavor and social message themes to harm content messages decreased the PME • Liking and interest: Adding flavor theme to harm and addiction content were correlated with higher likability scores	n/a
Xu et al., (2023)	Message content: (1) VR-self (2) VR-other (3) Print	• Liking and interest: vaping interest • Attitudes: perceived harm of secondhand aerosol (SHA) • Decision-making: prevention intentions • Advising others to behave likewise: persuade others to quit vaping; persuade others to not vape nearby	Main effect of VR game: VR messages produced more desired outcomes than print messages on: • Liking and interest: 1) VR-other generated less vaping interest immediately; 2) Both VR-Self and VR-Other generated less vaping interest after 1 week. • Attitudes: VR-Other generated a higher level of perceived harm of secondhand harm than the print advertisement after 1 week.	n/a
Zhao et al., (2023)	Message framing: (1) Gain frame (2) Loss frame Message format: (1) Individual texts (2) Simulated exchange	• Negative affect • Positive affect • PME • Liking and interest: message liking • Attitudes: beliefs of e-cig harms; attitudes • Decision-making: intention to quit	Main effect of framing: Compared to gain frame, the loss frame generated stronger • Negative affect Main effect of message format: Simulated text exchanges produced more desired outcomes than individual text messages on: • Liking and interest: better liked • PME • Positive affect • Attitudes: beliefs about e-cigarettes	Message frame × message format In simulated exchange conditions, the gain frame outperformed the loss frame for: • Liking and interest • Positive affect • Attitudes • Decision-making: intention to use The reverse trend was observed in individual text conditions.

aimed at youth and young adults. Six studies have examined the effects of gain versus loss framing on various communication outcomes (Cai & Zhao, 2021; Calabro et al., 2019; Jun et al., 2021; Kong et al., 2016; S. Liu & Yang, 2020b; Zhao et al., 2023). Of the four studies reporting the main effects of framing, conflicting results emerged. Two studies found an overall advantage of loss framing over gain framing (Cai & Zhao, 2021; Zhao et al., 2023). Specifically, Cai & Zhao (2021) revealed that the loss frame elicited more negative affective responses, stronger risk beliefs of vaping, less intention to vape, and higher PME than the gain frame. Similarly, Zhao et al. (2023) found that, compared to the gain frame, the loss frame generated stronger negative affect but no other significant differences were observed. In contrast, Calabro et al. (2019) reported that the gain frame elicited more risk beliefs of vaping, while Jun et al. (2021) did not find significant framing effects in their study.

Exploring interaction effects between framing and audience characteristics provided additional insights. Cai and Zhao (2021) found that the audience's uncertainty tolerance moderated the framing effect. In particular, gain frames proved more effective for those with low uncertainty tolerance, while loss frames were more effective for those with high uncertainty tolerance regarding PME, negative beliefs of vaping, attitudes toward vaping, and intention to vape. Kong et al. (2016) suggested that framing might be contingent on the demographics and vaping status of the audience. The study found that females preferred loss-framed

messages for all themes relative to males; lifetime e-cigarette users preferred loss-framed health risks and social labeling messages relative to never users; high school students preferred gain-framed social labeling messages relative to college students. Jun et al. (2021), however, reported that vaping status (vapers vs. non-vapers) did not moderate the effects of message framing on message reactions. On the contrary, Jun et al. (2021) found a three-way interaction between framing, audience vaping status, and affirmation: vapers demonstrated the highest level of support for vape-free policy when they were self-affirmed and exposed to a gain-framed message, while non-vapers exhibited the highest level of policy support when they were self-affirmed and exposed to a loss-framed message.

Additionally, three studies investigated the interaction of framing with other message characteristics. S. Liu and Yang (2020a) reported that gain-framed narratives, compared to non-narratives, resulted in lower intention to use e-cigarettes through guilt and increased risk perceptions of e-cigarettes, whereas loss-framed narratives decreased intention to vape through heightened sadness and increased risk perception, than loss-framed non-narratives. Zhao et al. (2023) examined how message format, either individual vaping risk text from a doctor or a simulated exchange of texts between a doctor and a youth, might moderate framing effects. The study found that the gain frame outperformed the loss frame for message liking, positive affect, attitudes toward vaping, and intention to vape in simulated exchange conditions, while in individual text

conditions, the reverse pattern was observed (Zhao et al., 2023). Lastly, findings from Kong et al. (2016) suggested that framing might be contingent on the anti-vaping message theme. For instance, loss frames were preferred over gain frames for themes related to health risks, addiction potential, and social labeling as a smoker. On the contrary, gain frames were preferred over loss frames for the theme related to financial cost.

Theme. Themes are also widely examined in the 25 reviewed studies. Boynton et al. (2022) conducted a systematic analysis of 220 vaping prevention ads to explore how coded message elements, including themes, could impact adolescents' perceptions of message effectiveness and vaping appeal. Youth participants were asked to rate the perceived effectiveness and appeal of six vaping prevention ads. The study discovered that certain themes, such as nicotine addiction, chemicals, negative health effects, health-related symptoms, and comparison to cigarettes, were associated with higher PME. Conversely, themes related to industry targeting, environmental impact, and flavors were linked to lower PME.

Several studies have underscored the heightened effectiveness of physical health messages (e.g., lung damage, brain messages) when compared to themes centered around nicotine and other topics. For example, Lazard (2021) compared lung damage themes to mood change, nicotine addiction, and vape design, revealing an overall benefit of lung damage themes on PME and cognitive elaboration. In another study, the nicotine theme was perceived as less effective compared to themes on chemical, lung, and COVID-19 harms (Rohde et al., 2022). Patterson et al. (2022) found that health harm messages were associated with lower intention to vape compared to wellness and pride themes among sexual minority individuals. Supporting these findings, Villanti et al. (2021) demonstrated that harm content messages generated higher PME than addiction content messages. In a longitudinal study where participants were exposed to text messages over six days, Noar et al. (2019) found that chemical and brain text messages induced more fear responses and higher PME than nicotine messages, despite the latter being better recalled and recognized. However, another longitudinal study reported no significant difference between health harm-themed and addiction-themed Real Cost video campaigns in terms of susceptibility and secondary outcomes to vaping at week 3 (Noar et al., 2022).

When exploring less common themes, Cartujano-Barrera et al. (2022) discovered that health rewards, financial rewards, autonomy, and social norms did not reduce susceptibility to vaping among Black-Hispanic youth participants.

Mixed findings emerged from three studies regarding the moderating effect of vaping status on message themes. For example, Boynton et al. (2022) found that the theme on lung effects was effective among e-cigarette users but not non-users. Additionally, themes on death were effective among users and susceptible but not non-susceptible non-users. Moreover, the word "addiction," themes on brain effects, and non-harmless water vapor were effective among susceptible and non-susceptible non-users but not for e-cigarette users. In contrast, Lazard (2021) did not find interaction effects of e-cigarette use

status and the effects of message themes. Same as Rohde et al. (2022), which also reported that the warning message theme had no interaction effect with vaping status. Beyond vaping status, Lee et al. (2022) demonstrated that sexual minority identity affirmation on PME was more pronounced with pride-themed messages. Conversely, Patterson et al. (2022) found no modification of the effect of message themes on any outcomes by sexual identity.

Dogmatic vs. suggestive language. Clayton et al. (2020) focused on the effects of dogmatic versus suggestive language in anti-vaping messages on psychological reactance and cognitive resources among young adult ever-vapers. The results indicated that exposure to dogmatic anti-vaping PSAs led to greater perceived freedom threats, anger, counterarguments, and intentions to vape compared to exposure to suggestive anti-vaping PSAs. Additionally, exposure to dogmatic anti-vaping PSAs was associated with slower response times and reduced encoding of content relative to suggestive anti-vaping PSAs.

Narrative vs. non-narrative language. S. Liu and Yang (2020b) compared the effects of narrative and non-narrative messages on young adults' attitudes and behavioral intentions toward e-cigarette use. The findings indicated that compared to young adults exposed to non-narrative messages, those who were exposed to narrative messages experienced greater transportation. However, transportation only influenced behavioral intention, not attitude. Additionally, the non-narrative message, which featured statistics and scientific evidence, appeared to trigger attitude change more readily than the narrative message. S. Liu and Yang (2020b) also highlighted that narrative persuasion seemed more effective in reinforcing existing negative attitudes toward e-cigarettes among non-users. Participants who have not used e-cigs before report more negative attitudes after reading the narrative message than those who have used e-cigarettes.

Message format. Lazard (2021) examined the impact of three message formats (visual-based messages, quizzes, and text-only messages) on message reactions, knowledge, beliefs, and sharing preferences among adolescent participants. The study found that all three message formats led to greater knowledge and beliefs about the harms of e-cigarettes compared to the control group, regardless of format. However, there was no significant difference in PME among the three message formats. The study also found that adolescents preferred visual-based messages and quizzes over text-only messages. Zhao et al. (2023) compared between simulated text exchanges and individual text messages, which found that simulated text exchanges are better liked, producing more positive affect, risk beliefs of e-cigarettes, and higher PME than individual texts. Lastly, Xu et al. (2023) explored the use of Virtual Reality (VR) messages in increasing risk perceptions and promoting preventive behaviors against vaping and secondhand e-cigarette aerosol. Participants were randomly assigned to view one of three messages: a VR message presenting the impact of secondhand aerosol on the self (VR-Self), a VR message showing the impact on others (VR-

Other), and a print advertisement. The study found that VR messages produced more desired outcomes than print messages. Specifically, VR-other generated less vaping interest immediately and both VR-Self and VR-Other generated less vaping interest after 1 week. In addition, VR-Other generated a higher level of perceived harm of secondhand harm than the print advertisement after 1 week.

Other message features. Images with vapor, devices, or e-liquid were not well-liked but perceived with higher PME in discouraging vaping (Stevens et al., 2021). A similar pattern was observed with images from warning labels, where warning label images were not well-liked but perceived as more effective in dissuading vaping than images not from a warning (Stevens et al., 2021). Sanders-Jackson et al. (2019) suggested that the presence of vapor activates the approach/appetitive motivational system, leading to greater positive affect, cognitive elaboration, memorization, but also higher urge to vape. Flavor content has been found to decrease message effectiveness. Adding flavor and social message themes to harm content messages decreased PME, although adding flavor to harm and addiction content increased the message likability scores (Villanti et al., 2021).

Channel

Three longitudinal studies utilized text messaging to transmit anti-vaping messages (Calabro et al., 2019; Graham et al., 2020; Noar et al., 2019). Although none of these studies directly compared various channels for message transmission, they collectively presented evidence indicating the receptiveness of youth and young adults to text messaging in the context of anti-vaping efforts. Calabro et al. (2019) reported substantial increases in knowledge and risk perceptions of e-cigarettes following exposure to text messages. In the study by Noar et al. (2019), posttest results indicated that knowledge about the harms of e-cigarettes, consideration of the risks associated with e-cigarettes, and perceived risks of e-cigarettes were all significantly higher compared to pretest measures. Graham et al. (2020) further demonstrated that participants receiving the text-message-based intervention were one-third more likely to quit vaping at the 7-month primary endpoint compared to control participants.

Audience

Vaping status. Vaping status plays a crucial role in shaping the reactions and perceptions of anti-vaping messages among youth and young adults. For instance, Jun et al. (2021) found that non-vapers expressed higher levels of policy support compared to vapers. Rath et al. (2021) further explained this phenomenon, revealing that non-susceptible never-users exhibited the highest mean scale scores on risk beliefs of e-cigarettes, followed by susceptible never-users, ever users, and current users in decreasing order.

Uncertainty tolerance. In addition to the discussed interaction between framing and uncertainty tolerance on message effectiveness, Cai and Zhao (2021) demonstrated that lower

uncertainty tolerance yielded more favorable outcomes in terms of PME, negative affect, positive affect, and risk beliefs of vaping compared to higher uncertainty tolerance.

Audience affirmation. Jun et al. (2021) also explored the role of audience affirmation and found that self-affirmation was associated with a higher level of support for vape-free policies compared to group-affirmation.

SGM identity. Lee et al. (2022) suggested that affirmation of a few sexual and gender minority (SGM) identities was significantly and positively associated with PME with This Free Life, a national anti-tobacco campaign targeting SGM youth and young adults.

Social media and internet use. J. Liu et al. (2023) investigated the association between social media/internet use and young adults' intentions to share anti-vaping social media posts. The findings indicated that the total number of social media sites used was associated with intentions to "like" the posts and the overall engagement score. Additionally, daily internet use was linked to intentions to "comment on" and "like" the posts.

Discussion

The primary objective of this systematic review was to conduct a thorough analysis and synthesis of existing literature, encompassing 25 identified studies that specifically address messaging strategies in anti-vaping communication aimed at youth and young adults. The coding process involved categorizing articles based on key study characteristics, including research method, study sample, and study design. Additionally, articles were classified according to three types of theories – behavior change, message effect, and human information processing theories – following the framework proposed by Cappella (2006). Building upon McGuire's Communication/Persuasion model (McGuire, 2013), the 25 studies were further coded according to communication inputs and outputs. Finally, the findings of studies on both the main and interaction effects of various communication inputs on outputs were reviewed. National anti-vaping campaigns and an intervention appeared in this review, including The Real Cost E-Cigarette Prevention campaign (Noar et al., 2022), This Free Life campaign (Lee et al., 2022) and This Is Quitting intervention (Patterson et al., 2022).

The systematic review underscores the considerable research focus on anti-vaping communication, a critical area given the urgent public health concerns surrounding vaping among youth and young adults. In the subsequent sections, we will discuss the primary findings and major gaps identified in the 25 reviewed studies across study characteristics, theoretical frameworks, and the effects of five examined communication inputs.

Study characteristics

The extensive use of diverse research methods, including surveys, randomized clinical trials, and laboratory experiments, alongside both cross-sectional and longitudinal designs,

demonstrate the wide range of research approaches employed in the reviewed studies. However, a notable gap exists as all reviewed studies were conducted in the US, which may limit the generalizability of the findings to other cultural or geographical contexts. Hence, further research is warranted to broaden the scope and enhance the applicability of the vaping messaging strategies.

Furthermore, the wide range of communication outputs examined in this review emphasizes the need to understand various persuasive outcome measurements and their distinct effectiveness in predicting behavioral changes. For studies assessing more than one persuasive outcome, it is essential to identify the most indicative metrics of behavioral changes, and clearly distinguish primary from secondary outcomes to inform future message and intervention design (Noar et al., 2022). Further research is needed to explore the nuanced dynamics of diverse persuasive outcome measurements, providing clear guidance for researchers and practitioners in the design and selection of anti-vaping messages.

Theoretical frameworks

While 17 out of 25 studies incorporated theories to inform the design and/or evaluation of message effectiveness, the utilization of additional theories could provide deeper insights into the mechanisms underlying message effects and enhance understanding of the effectiveness of diverse messaging strategies. For instance, future research could examine various message features using theories such as the emotions-as-frames model (Nabi, 2003, 2007), explore the underlying mechanisms of message efficacy through human information processing theories like the elaboration likelihood model (Petty & Cacioppo, 1986), and investigate individual characteristics that influence the effectiveness of anti-vaping messages using behavior change theories such as the transtheoretical model (Prochaska & Velicer, 1997).

Furthermore, integrating theories from all three categories can provide a comprehensive understanding of anti-vaping messages among specific audience groups, shedding light on psychological and cognitive pathways. Only three reviewed studies integrated theories from multiple categories. Future research should aim to incorporate and examine theories from all three categories collectively to enrich our understanding of anti-vaping messaging strategies targeting youth and young adults.

Source

The findings on communication inputs underscore the importance of considering message sources in anti-vaping messages. Although one study suggested that young adults viewed expert messages as more credible than peer messages, leading to increased perceptions of e-cigarette harm (Lee & Stevens, 2022), the perceptions of youth regarding various message sources such as social media influencers in vaping prevention and cessation remain unclear. In addition, further research is needed to address this gap and gain valuable insights into how youth and young adults perceive different message sources related to vaping, considering not only credibility but also

other source characteristics widely employed in health communication, such as similarity (Silvia, 2005), attractiveness, and likability (Tucker Smith & De Houwer, 2014).

Message

Interestingly, the theme of nicotine addiction, commonly used in current vaping health messages (Kresovich et al., 2022), was found to be less persuasive compared to themes like harmful chemicals, lung harm, and brain effects (Noar et al., 2019; Patterson et al., 2022; Rohde et al., 2022; Villanti et al., 2021). Research has suggested that youth may inaccurately compare nicotine addiction to other forms of addiction, such as fast-food addiction, which could lead to a minimization of perceived risk (Roditis et al., 2020). On the contrary, youth were particularly alarmed and frightened when familiar chemical terms (e.g., carbon monoxide, etc.) were used, while the use of uncommon chemical terms (e.g., acrolein) may cause confusion (Popova et al., 2021). Therefore, to maximize message persuasiveness, further research should investigate not only the inclusion of different themes but also how these themes are explained to youth.

The findings from the systematic review present a complex picture of the moderating effect of message themes and audience characteristics. For instance, Boynton et al. (2022) found that themes of lung effects and death were effective among e-cigarette users but not non-users; On the contrary, addiction and brain effects, as well as non-harmless water vapor themes were effective among non-users but not among users. However, two studies did not find any interaction effects of e-cigarette use status and the effects of message themes (Lazard, 2021; Rohde et al., 2022). Besides vaping status, Lee et al. (2022) demonstrated that sexual minority identity affirmation moderated the effect of a pride-theme anti-vaping campaign. Further research is essential to fully understand the effectiveness of various themes and whether they demonstrate similar efficacy across diverse audience groups, considering not only vaping status but also social-demographic variables, personality traits, and diverse experiences.

Compared to the extensive research on the gain vs. loss-frame in cigarette-cessation-related messages (e.g., Cornacchione & Smith, 2012; Lipkus et al., 2013; Toll et al., 2010), no research in this review specifically investigated message framing in vaping-cessation-related health messages for youth and young adults. Furthermore, the conflicting results and intricate interplay of framing with audience characteristics and other message elements in vaping prevention messages highlight the necessity for additional research. Future studies could explore in greater depth the nuances of how framing interacts with diverse audience characteristics, such as age, gender, cultural background, and prior vaping experience.

Language styles of anti-vaping messages can significantly influence their effectiveness among youth and young adults (Clayton et al., 2020; S. Liu & Yang, 2020b). Therefore, future research should continue to explore the effects of different language styles, including but not limited to dogmatic, suggestive, narrative, and non-narrative language. The reviewed studies also highlight the need to consider audience characteristics when considering language style for anti-

vaping messages. For instance, narrative messages may be particularly effective for non-users, while suggestive language may be more effective for current vapers. Furthermore, the inclusion of other message features, such as vapor, should be approached with caution due to mixed results regarding its effectiveness. Specifically, the arousing feature of vapor has been shown to increase the audience's urge to vape (Boynton et al., 2022; Sanders-Jackson et al., 2019; Stevens et al., 2021; Tan et al., 2018).

Audience

While four studies targeted specific sub-groups within the youth and young adult population, such as Black and Latino youth (Cartujano-Barrera et al., 2022), SGM young adults (Lee et al., 2022; Patterson et al., 2022), and current smokers (Tan et al., 2018), additional risk factors for vaping among youth and young adults, such as mental health issues (Becker et al., 2021), substance use disorders (Chadi et al., 2019), and disability (Zhang et al., 2022), underscore the necessity for targeted communication research among various populations to address vaping-related public health concerns effectively.

Tailored health communication, which considers individuals' unique needs, preferences, and backgrounds, has gained recognition in health message development (Noar et al., 2009). Tailoring messages based on audience characteristics such as race, gender, age, and quitting status has been associated with increased persuasive effects of anti-smoking messages (Kim et al., 2016; Strecher et al., 2008). Diverse "peer crowds," characterized by shared values, cultural interests, and personal styles, also exert significant influence on tobacco-related attitudes and behaviors, highlighting the critical need for culturally tailored research (Jordan et al., 2019; Moran et al., 2017). Future research should also investigate the application of the above tailoring strategies in health communication aimed at vaping prevention and cessation among youth and young adults.

Channel

The findings from reviewed studies also have significant implications for the design and implementation of future vaping campaigns and interventions. Lazard (2021) shows that adolescents prefer visual-based messages and quizzes over text-only messages, indicating that incorporating more visual elements and interactive content like quizzes could potentially enhance the engagement and effectiveness of anti-vaping social media campaigns. Zhao et al. (2023) found that simulated text exchanges are more effective than individual text messages in generating positive communication outcomes. This finding is particularly relevant, given the adoption of text messages in disseminating vaping prevention and cessation interventions (Graham et al., 2021; Noar et al., 2019). Lastly, the study by Xu et al. (2023) demonstrated the potential of VR messages in increasing risk perceptions and promoting preventive behaviors against vaping and secondhand e-cigarette aerosol.

Although numerous longitudinal studies have demonstrated the receptiveness of youth and young adults to anti-

vaping text messages (Graham et al., 2020; Noar et al., 2019), future research should continue to explore and evaluate diverse message channels, such as social media to identify the most effective platforms among various target audiences. Additionally, as technology continues to evolve, it will be important to stay abreast of innovative communication technologies such as VR that could be leveraged to enhance the reach and impact of anti-vaping messages.

Destination

Notably, only one intervention study focused on vaping cessation. Quitting vaping can be challenging for youth and young adults due to factors such as the high addictive potential of vaping (Jankowski et al., 2019), the diversity of vaping products, ease of use, and the social acceptance of vaping among peers (Sanchez et al., 2021). In the 2021 NYTS, 68% of youth who currently vape reported unsuccessful attempts to quit vaping within the previous year (Dai et al., 2023). Despite these challenges, there is a lack of campaign initiatives aimed at promoting vaping cessation and encouraging the use of quitting resources among youth and young adults currently using e-cigarettes. Existing national cessation campaigns like CDC's Tips from Former Smokers and FDA's Every Try Counts have primarily targeted adults and cigarette smoking. Research shows that the Tips campaign has contributed to 16.4 million quit attempts and over 1 million estimated sustained quits among adults from 2012 to 2018 (Murphy-Hoefer et al., 2020). Therefore, further research is warranted to explore messaging strategies that raise awareness about quitting, promote vaping resources, and encourage abstinence among youth and young adults using e-cigarettes and dual users of e-cigarettes and cigarettes (Tan et al., 2018).

Conclusion

This systematic review sheds light on the effects and nuanced interplay between various communication inputs, encompassing message themes, framing, language style, message format, message source, disseminating channels, and audience characteristics. Nevertheless, the predominance of U.S.-based studies and the limited research aimed at promoting vaping cessation and encouraging the use of quitting resources highlights the necessity for more inclusive research and communication efforts. Drawing from McGuire's Communication/Persuasion Model, this review underscores the need for targeted and tailored public health communication to effectively address vaping prevention and cessation among youth and young adults, considering the impact and interactions of message features, sources, channels, and audience characteristics.

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References

- Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes*, 50(2), 179–211. [https://doi.org/10.1016/0749-5978\(91\)90020-T](https://doi.org/10.1016/0749-5978(91)90020-T)
- Allen, J. A., Duke, J. C., Davis, K. C., Kim, A. E., Nonnemaker, J. M., & Farrelly, M. C. (2015). Using mass media campaigns to reduce youth tobacco use: A review. *American Journal of Health Promotion*, 30(2), e71–e82. <https://doi.org/10.4278/ajhp.130510-LIT-237>
- Andrews, J. C., Mays, D., Netemeyer, R. G., Burton, S., & Kees, J. (2019). Effects of e-cigarette health warnings and modified risk ad claims on adolescent e-cigarette craving and susceptibility. *Nicotine & Tobacco Research*, 21(6), 792–798. <https://doi.org/10.1093/ntr/nty076>
- Arnold, J., Bailey, C. P., Evans, W. D., & Napolitano, M. A. (2022). Application of McGuire's model to weight management messages: Measuring persuasion of Facebook posts in the healthy body, healthy U trial for young adults attending university in the United States. *International Journal of Environmental Research and Public Health*, 19(21), 14275. <https://doi.org/10.3390/ijerph192114275>
- Becker, T. D., Arnold, M. K., Ro, V., Martin, L., & Rice, T. R. (2021). Systematic review of electronic cigarette use (vaping) and mental health comorbidity among adolescents and young adults. *Nicotine & Tobacco Research*, 23(3), 415–425. <https://doi.org/10.1093/ntr/ntaa171>
- Birdsey, J., Cornelius, M., Jamal, A., Park-Lee, E., Cooper, M. R., Wang, J., Sawdey, M. D., Cullen, K. A., & Neff, L. (2023). Tobacco product use among U.S. middle and high school students - National youth tobacco survey, 2023. *MMWR Morbidity and Mortality Weekly Report*, 72(44), 1173–1182. <https://doi.org/10.15585/mmwr.mm7244a1>
- Boynton, M. H., Sanzo, N., Brothers, W., Kresovich, A., Sutfin, E. L., Sheeran, P., & Noar, S. M. (2022). Perceived effectiveness of objective elements of vaping prevention messages among adolescents. *Tobacco Control*, 32(e2), e228–e235. <https://doi.org/10.1136/tobaccocontrol-2021-057151>
- Cai, X., & Zhao, X. (2021). Framing youth vaping prevention messages: The role of uncertainty tolerance. *Health Communication*, 38(4), 670–680. <https://doi.org/10.1080/10410236.2021.1966181>
- Calabro, K. S., Khalil, G. E., Chen, M., Perry, C. L., & Prokhorov, A. V. (2019). Pilot study to inform young adults about the risks of electronic cigarettes through text messaging. *Addictive Behaviors Reports*, 10, 100224. <https://doi.org/10.1016/j.abrep.2019.100224>
- Cappella, J. N. (2006). Integrating message effects and behavior change theories: Organizing comments and unanswered questions. *Journal of Communication*, 56(suppl_1), S265–S279. <https://doi.org/10.1111/j.1460-2466.2006.00293.x>
- Cartujano-Barrera, F., Hernández-Torrez, R., Cai, X., Orfin, R. H., Azogini, C., Chávez-Iñiguez, A., Santa Cruz, E., Bansal-Travers, M., Wilson, K. M., McIntosh, S., Ossip, D. J., & Cupertino, A. P. (2022). Evaluating the immediate impact of graphic messages for vaping prevention among black and latino adolescents: A randomized controlled trial. *International Journal of Environmental Research and Public Health*, 19(16), 10026. <https://doi.org/10.3390/ijerph191610026>
- Cavallo, D. A., Kong, G., Ells, D. M., Camenga, D. R., Morean, M. E., & Krishnan-Sarin, S. (2019). Youth generated prevention messages about electronic cigarettes. *Health Education Research*, 34(2), 247–256. <https://doi.org/10.1093/her/cyz001>
- Chadi, N., Schroeder, R., Jensen, J. W., & Levy, S. (2019). Association between electronic cigarette use and marijuana use among adolescents and young adults: A systematic review and meta-analysis. *JAMA Pediatrics*, 173(10), e192574. <https://doi.org/10.1001/jamapediatrics.2019.2574>
- Chan, G. C. K., Gartner, C., Lim, C., Sun, T., Hall, W., Connor, J., Stjepanović, D., & Leung, J. (2022). Association between the implementation of tobacco control policies and adolescent vaping in 44 lower-middle, upper-middle, and high-income countries. *Addiction*, 117(8), 2296–2305. <https://doi.org/10.1111/add.15892>
- Clayton, R. B., Lang, A., Leshner, G., & Quick, B. L. (2019). Who fights, who flees? An integration of the LC4MP and psychological reactance theory. *Media Psychology*, 22(4), 545–571. <https://doi.org/10.1080/15213269.2018.1476157>
- Clayton, R. B., Leshner, G., Sanders-Jackson, A., & Hendrickse, J. (2020). When countering becomes the primary task: Examination of dogmatic anti-vaping messages on psychological reactance, available cognitive resources, and memory. *Journal of Communication*, 70(4), 522–547. <https://doi.org/10.1093/joc/jqaa010>
- Cornacchione, J., & Smith, S. W. (2012). The effects of message framing within the stages of change on smoking cessation intentions and behaviors. *Health Communication*, 27(6), 612–622. <https://doi.org/10.1080/10410236.2011.619252>
- Dai, H. D., Hanh, P., Guenzel, N., Morgan, M., Kerns, E., & Winickoff, J. P. (2023). Adoption of vaping cessation methods by US adolescent e-cigarette users. *Pediatrics*, 152(5), e2023062948. <https://doi.org/10.1542/peds.2023-062948>
- England, L. J., Bunnell, R. E., Pechacek, T. F., Tong, V. T., & McAfee, T. A. (2015). Nicotine and the developing human: A neglected element in the electronic cigarette debate. *American Journal of Preventive Medicine*, 49(2), 286–293. <https://doi.org/10.1016/j.amepre.2015.01.015>
- Farzal, Z., Perry, M. F., Yarbrough, W. G., & Kimple, A. J. (2019). The adolescent vaping epidemic in the United States—How it happened and where we go from here. *JAMA Otolaryngology-Head & Neck Surgery*, 145(10), 885–886. <https://doi.org/10.1001/jamaoto.2019.2410>
- Graham, A. L., Amato, M. S., Cha, S., Jacobs, M. A., Bottcher, M. M., & Papandonatos, G. D. (2021). Effectiveness of a vaping cessation text message program among young adult e-cigarette users: A randomized clinical trial. *JAMA Internal Medicine*, 181(7), 923–930. <https://doi.org/10.1001/jamainternmed.2021.1793>
- Graham, A. L., Jacobs, M. A., Amato, M. S., Cha, S., Bottcher, M. M., & Papandonatos, G. D. (2020). Effectiveness of a quit vaping text message program in promoting abstinence among young adult e-cigarette users: Protocol for a randomized controlled trial. *JMIR Research Protocols*, 9(5), e18327. <https://doi.org/10.2196/18327>
- Hammond, D., Reid, J. L., Cole, A. G., & Leatherdale, S. T. (2017). Electronic cigarette use and smoking initiation among youth: A longitudinal cohort study. *Canadian Medical Association Journal*, 189(43), E1328–E1336. <https://doi.org/10.1503/cmaj.161002>
- Jankowski, M., Krzystanek, M., Zejda, J. E., Majek, P., Lubanski, J., Lawson, J. A., & Brozek, G. (2019). E-cigarettes are more addictive than traditional cigarettes—A study in highly educated young people. *International Journal of Environmental Research and Public Health*, 16(13), 2279. <https://doi.org/10.3390/ijerph16132279>
- Jin, L., Lynch, J., Richardson, A., Lorkiewicz, P., Srivastava, S., Theis, W., Shirk, G., Hand, A., Bhatnagar, A., Srivastava, S., & Conklin, D. J. (2021). Electronic cigarette solvents, pulmonary irritation, and endothelial dysfunction: Role of acetaldehyde and formaldehyde. *American Journal of Physiology-Heart and Circulatory Physiology*, 320(4), H1510–H1525. <https://doi.org/10.1152/ajpheart.00878.2020>
- Jordan, J. W., Stalgaitis, C. A., Charles, J., Madden, P. A., Radhakrishnan, A. G., & Saggese, D. (2019). Peer crowd identification and adolescent health behaviors: Results from a statewide representative study. *Health Education & Behavior*, 46(1), 40–52. <https://doi.org/10.1177/1090198118759148>
- Jun, J., Wen, T. J., & Wu, L. (2021). The effects of self vs. group affirmation and message framing on college students' vape-free campus policy support. *Health Communication*, 36(11), 1441–1451. <https://doi.org/10.1080/10410236.2020.1767446>
- Kidd, L. R., Garrard, G. E., Bekessy, S. A., Mills, M., Camilleri, A. R., Fidler, F., Fielding, K. S., Gordon, A., Gregg, E. A., Kusmanoff, A. M.,

- Louis, W., Moon, K., Robinson, J. A., Selinske, M. J., Shanahan, D., & Adams, V. M. (2019). Messaging matters: A systematic review of the conservation messaging literature. *Biological Conservation*, 236, 92–99. <https://doi.org/10.1016/j.biocon.2019.05.020>
- Kim, M. (2019). When similarity strikes back: Conditional persuasive effects of character-audience similarity in anti-smoking campaign. *Human Communication Research*, 45(1), 52–77. <https://doi.org/10.1093/hcr/hqy013>
- Kim, M., Shi, R., & Cappella, J. N. (2016). Effect of character–audience similarity on the perceived effectiveness of antismoking PSAs via engagement. *Health Communication*, 31(10), 1193–1204. <https://doi.org/10.1080/10410236.2015.1048421>
- Kong, G., Cavallo, D. A., Camenga, D. R., Morean, M. E., & Krishnan-Sarin, S. (2016). Preference for gain- or loss-framed electronic cigarette prevention messages. *Addictive Behaviors*, 62, 108–113. <https://doi.org/10.1016/j.addbeh.2016.06.015>
- Kowitz, S. D., Mendel Sheldon, J., Vereen, R. N., Kurtzman, R. T., Gottfredson, N. C., Hall, M. G., Brewer, N. T., & Noar, S. M. (2023). The impact of the real cost vaping and smoking ads across tobacco products. *Nicotine & Tobacco Research*, 25(3), 430–437. <https://doi.org/10.1093/ntr/ntac206>
- Kramarow, E. A., & Elgaddal, N. (2023). Current electronic cigarette use among adults aged 18 and over: United States, 2021. *NCHS Data Brief*, 475, 1–8. <https://doi.org/10.15620/cdc.129966>
- Kresovich, A., Sanzo, N., Brothers, W., Prentice-Dunn, H., Boynton, M. H., Sutfin, E. L., Sheeran, P., & Noar, S. M. (2022). What's in the message? An analysis of themes and features used in vaping prevention messages. *Addictive Behaviors Reports*, 15, 100404. <https://doi.org/10.1016/j.abrep.2021.100404>
- Lang, A. (2000). The limited capacity model of mediated message processing. *Journal of Communication*, 50(1), 46–70. <https://doi.org/10.1111/j.1460-2466.2000.tb02833.x>
- Lazard, A. J. (2021). Social media message designs to educate adolescents about e-cigarettes. *Journal of Adolescent Health*, 68(1), 130–137. <https://doi.org/10.1016/j.jadohealth.2020.05.030>
- Lazaro, A., Ceballos, R., Fischer, M., Smuin, S., & Halpern-Felsher, B. (2021). A novel approach to training educators to conduct school-based adolescent e-cigarette education and prevention: Using the tobacco prevention toolkit. *Addictive Behaviors*, 118, 106858. <https://doi.org/10.1016/j.addbeh.2021.106858>
- Lee, D. N., & Stevens, E. M. (2022). Message source credibility and e-cigarette harm perceptions among young adults. *International Journal of Environmental Research and Public Health*, 19(15), 9123. <https://doi.org/10.3390/ijerph19159123>
- Lee, D. N., Stevens, E. M., Keller-Hamilton, B., Wedel, A. V., Wagener, T. L., & Patterson, J. G. (2022). Minoritized sexual identity and perceived effectiveness of Instagram public health messaging about e-cigarettes. *Journal of Health Communication*, 27(2), 115–124. <https://doi.org/10.1080/10810730.2022.2059724>
- Lipkus, I. M., Ranby, K. W., Lewis, M. A., & Toll, B. (2013). Reactions to framing of cessation messages: Insights from dual-smoker couples. *Nicotine & Tobacco Research*, 15(12), 2022–2028. <https://doi.org/10.1093/ntr/ntt091>
- Liu, J., Lee, D. N., & Stevens, E. M. (2023). Characteristics associated with young adults' intentions to engage with anti-vaping Instagram posts. *International Journal of Environmental Research and Public Health*, 20(11), 6054. <https://doi.org/10.3390/ijerph20116054>
- Liu, S., & Yang, J. Z. (2020a). Incorporating message framing into narrative persuasion to curb e-cigarette use among college students. *Risk Analysis*, 40(8), 1677–1690. <https://doi.org/10.1111/risa.13502>
- Liu, S., & Yang, J. Z. (2020b). The role of temporal distance perception in narrative vs. non-narrative persuasion related to e-cigarettes. *Journal of Health Communication*, 25(7), 543–553. <https://doi.org/10.1080/10810730.2020.1788678>
- Lyu, J. C., Olson, S. S., Ramo, D. E., & Ling, P. M. (2022). Delivering vaping cessation interventions to adolescents and young adults on Instagram: Protocol for a randomized controlled trial. *BMC Public Health*, 22(1), 2311. <https://doi.org/10.1186/s12889-022-14606-7>
- Ma, H., Kieu, T. K.-T., Ribisl, K. M., & Noar, S. M. (2023). Do vaping prevention messages impact adolescents and young adults? A meta-analysis of experimental studies. *Health Communication*, 38(8), 1709–1722. <https://doi.org/10.1080/10410236.2023.2185578>
- McGuire, W. J. (1984). Public communication as a strategy for inducing health-promoting behavioral change. *Preventive Medicine*, 13(3), 299–319. [https://doi.org/10.1016/0091-7435\(84\)90086-0](https://doi.org/10.1016/0091-7435(84)90086-0)
- McGuire, W. J. (2013). McGuire's classic input-output framework for constructing persuasive messages. In R. E. Rice & C. K. Atkin (Eds.), *Public communication campaigns* (4th ed., pp. 133–145). SAGE Publications, Inc.
- Moran, M. B., Walker, M. W., Alexander, T. N., Jordan, J. W., & Wagner, D. E. (2017). Why peer crowds matter: Incorporating youth subcultures and values in health education campaigns. *American Journal of Public Health*, 107(3), 389–395. <https://doi.org/10.2105/AJPH.2016.303595>
- Murphy-Hoefer, R., Davis, K. C., King, B. A., Beistle, D., Rodes, R., & Graffunder, C. (2020). Association between the tips from former smokers campaign and smoking cessation among adults, United States, 2012–2018. *Preventing Chronic Disease*, 17, E97. <https://doi.org/10.5888/pcd17.200052>
- Nabi, R. L. (2003). Exploring the framing effects of emotion: Do discrete emotions differentially influence information accessibility, information seeking, and policy preference? *Communication Research*, 30(2), 224–247. <https://doi.org/10.1177/0093650202250881>
- Nabi, R. L. (2007). Emotion and persuasion: A social cognitive perspective. In D. R. Roskos-Ewoldsen & J. L. Monahan (Eds.), *Communication and social cognition: Theories and methods* (pp. 377–398). Lawrence Erlbaum Associates Publishers.
- Noar, S. M., Gottfredson, N. C., Kieu, T., Rohde, J. A., Hall, M. G., Ma, H., Fendinger, N. J., & Brewer, N. T. (2022). Impact of vaping prevention advertisements on US adolescents: A randomized clinical trial. *JAMA Network Open*, 5(10), e2236370. <https://doi.org/10.1001/jamanetworkopen.2022.36370>
- Noar, S. M., Gottfredson, N., Vereen, R. N., Kurtzman, R., Sheldon, J. M., Adams, E., Hall, M. G., & Brewer, N. T. (2021). Development of the UNC perceived message effectiveness scale for youth. *Tobacco Control*, 32(5), 553–558. <https://doi.org/10.1136/tobaccocontrol-2021-056929>
- Noar, S. M., Harrington, N. G., & Aldrich, R. S. (2009). The role of message tailoring in the development of persuasive health communication messages. *Annals of the International Communication Association*, 33(1), 73–133. <https://doi.org/10.1080/23808985.2009.11679085>
- Noar, S. M., Rohde, J. A., Horvitz, C., Lazard, A. J., Cornacchione Ross, J., & Sutfin, E. L. (2019). Adolescents' receptivity to e-cigarette harms messages delivered using text messaging. *Addictive Behaviors*, 91, 201–207. <https://doi.org/10.1016/j.addbeh.2018.05.025>
- O'Brien, D., Long, J., Quigley, J., Lee, C., McCarthy, A., & Kavanagh, P. (2021). Association between electronic cigarette use and tobacco cigarette smoking initiation in adolescents: A systematic review and meta-analysis. *BMC Public Health*, 21(1), 954. <https://doi.org/10.1186/s12889-021-10935-1>
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., & Moher, D. (2021). Updating guidance for reporting systematic reviews: Development of the PRISMA 2020 statement. *Journal of Clinical Epidemiology*, 134, 103–112. <https://doi.org/10.1016/j.jclinepi.2021.02.003>
- Patterson, J. G., Keller-Hamilton, B., Wedel, A. V., Wagener, T. L., & Stevens, E. M. (2022). Responses to e-cigarette health messages among young adult sexual minoritized women and nonbinary people assigned female at birth: Assessing the influence of message theme and format. *Drug & Alcohol Dependence*, 231, 109249. <https://doi.org/10.1016/j.drugalcdep.2021.109249>
- Petty, R. E., & Cacioppo, J. T. (1986). The elaboration likelihood model of persuasion. In L. Berkowitz (Ed.), *Advances in experimental social psychology* (Vol. 19, pp. 123–205). Academic Press. [https://doi.org/10.1016/S0065-2601\(08\)60214-2](https://doi.org/10.1016/S0065-2601(08)60214-2)
- Popova, L., Fairman, R. T., Akani, B., Dixon, K., & Weaver, S. R. (2021). "Don't do vape, bro!" a qualitative study of youth's and parents' reactions to e-cigarette prevention advertisements. *Addictive Behaviors*, 112, 106565. <https://doi.org/10.1016/j.addbeh.2020.106565>

- Prochaska, J. O., & Velicer, W. F. (1997). The transtheoretical model of health behavior change. *American Journal of Health Promotion*, 12(1), 38–48. <https://doi.org/10.4278/0890-1171-12.1.38>
- Rath, J. M., Romberg, A. R., Perks, S. N., Edwards, D., Vallone, D. M., & Hair, E. C. (2021). Identifying message themes to prevent e-cigarette use among youth and young adults. *Preventive Medicine*, 150, 106683. <https://doi.org/10.1016/j.ypmed.2021.106683>
- Riley, K. E., Ulrich, M. R., Hamann, H. A., & Ostroff, J. S. (2017). Decreasing smoking but increasing stigma? Anti-tobacco campaigns, public health, and cancer care. *AMA Journal of Ethics*, 19(5), 475–485. <https://doi.org/10.1001/journalofethics.2017.19.5.msoc1-1705>
- Roditis, M. L., Dineva, A., Smith, A., Walker, M., Delahanty, J., D'Lorio, E., & Holtz, K. D. (2020). Reactions to electronic nicotine delivery system (ENDS) prevention messages: Results from qualitative research used to inform FDA's first youth ENDS prevention campaign. *Tobacco Control*, 29(5), 510–515. <https://doi.org/10.1136/tobaccocontrol-2019-055104>
- Rohde, J. A., Noar, S. M., Sheldon, J. M., Hall, M. G., Kieu, T., & Brewer, N. T. (2022). Identifying promising themes for adolescent vaping warnings: A national experiment. *Nicotine & Tobacco Research*, 24(9), 1379–1385. <https://doi.org/10.1093/ntr/ntac093>
- Ross, C. J., Noar, S. M., & Sutfin, E. L. (2019). Systematic review of health communication for non-cigarette tobacco products. *Health Communication*, 34(3), 361–369. <https://doi.org/10.1080/10410236.2017.1407274>
- Rothman, A. J., & Salovey, P. (1997). Shaping perceptions to motivate healthy behavior: The role of message framing. *Psychological Bulletin*, 121(1), 3–19. <https://doi.org/10.1037/0033-2909.121.1.3>
- Sanchez, S., Kaufman, P., Pelletier, H., Baskerville, B., Feng, P., O'Connor, S., Schwartz, R., & Chaiton, M. (2021). Is vaping cessation like smoking cessation? A qualitative study exploring the responses of youth and young adults who vape e-cigarettes. *Addictive Behaviors*, 113, 106687. <https://doi.org/10.1016/j.addbeh.2020.106687>
- Sanders-Jackson, A., Clayton, R. B., Tan, A. S. L., & Yie, K. (2019). Testing the effect of vapor in ENDS public service announcements on current smokers and ends users' psychophysiological responses and smoking and vaping urge. *Journal of Health Communication*, 24(4), 413–421. <https://doi.org/10.1080/10810730.2019.1630525>
- Sapru, S., Vardhan, M., Li, Q., Guo, Y., Li, X., & Saxena, D. (2020). E-cigarettes use in the United States: Reasons for use, perceptions, and effects on health. *BMC Public Health*, 20(1), 1518. <https://doi.org/10.1186/s12889-020-09572-x>
- Silvia, P. J. (2005). Deflecting reactance: The role of similarity in increasing compliance and reducing resistance. *Basic and Applied Social Psychology*, 27(3), 277–284. https://doi.org/10.1207/s15324834baso2703_9
- Soneji, S., Barrington-Trimis, J. L., Wills, T. A., Leventhal, A. M., Unger, J. B., Gibson, L. A., Yang, J., Primack, B. A., Andrews, J. A., Miech, R. A., Spindle, T. R., Dick, D. M., Eissenberg, T., Hornik, R. C., Dang, R., & Sargent, J. D. (2017). Association between initial use of e-cigarettes and subsequent cigarette smoking among adolescents and young adults: A systematic review and meta-analysis. *JAMA Pediatrics*, 171(8), 788–797. <https://doi.org/10.1001/jamapediatrics.2017.1488>
- Stevens, E. M., Keller-Hamilton, B., Mays, D., Unger, J. B., Wackowski, O. A., West, J. C., & Villanti, A. C. (2021). Optimizing images for an e-cigarette messaging campaign: Liking and perceived effectiveness. *International Journal of Environmental Research and Public Health*, 18(24), 12989. <https://doi.org/10.3390/ijerph182412989>
- Strecher, V. J., McClure, J. B., Alexander, G. L., Chakraborty, B., Nair, V. N., Konkel, J. M., Greene, S. M., Collins, L. M., Carlier, C. C., Wiese, C. J., Little, R. J., Pomerleau, C. S., & Pomerleau, O. F. (2008). Web-based smoking-cessation programs: Results of a randomized trial. *American Journal of Preventive Medicine*, 34(5), 373–381. <https://doi.org/10.1016/j.amepre.2007.12.024>
- Tan, A. S. L., Rees, V. W., Rodgers, J., Agudile, E., Sokol, N. A., Yie, K., & Sanders-Jackson, A. (2018). Effects of exposure to anti-vaping public service announcements among current smokers and dual users of cigarettes and electronic nicotine delivery systems. *Drug & Alcohol Dependence*, 188, 251–258. <https://doi.org/10.1016/j.drugalcdep.2018.04.013>
- Tehrani, H., Rajabi, A., Ghelichi-Ghojogh, M., Nejatian, M., & Jafari, A. (2022). The prevalence of electronic cigarettes vaping globally: A systematic review and meta-analysis. *Archives of Public Health*, 80(1), 240. <https://doi.org/10.1186/s13690-022-00998-w>
- Toll, B. A., Martino, S., Latimer, A., Salovey, P., O'Malley, S., Carlin-Menter, S., Hopkins, J., Wu, R., Celestino, P., & Cummings, K. M. (2010). Randomized trial: Quitline specialist training in gain-framed vs standard-care messages for smoking cessation. *JNCI Journal of the National Cancer Institute*, 102(2), 96–106. <https://doi.org/10.1093/jnci/djp468>
- Tucker Smith, C., & De Houwer, J. (2014). The impact of persuasive messages on IAT performance is moderated by source attractiveness and likeability. *Social Psychology*, 45(6), 437–448. <https://doi.org/10.1027/1864-9335/a000208>
- U.S. Department of Health and Human Services. (2014). *The health consequences of smoking – 50 years of progress: A report of the surgeon general*.
- Villanti, A. C., LePine, S. E., West, J. C., Cruz, T. B., Stevens, E. M., Tetreault, H. J., Unger, J. B., Wackowski, O. A., & Mays, D. (2021). Identifying message content to reduce vaping: Results from online message testing trials in young adult tobacco users. *Addictive Behaviors*, 115, 106778. <https://doi.org/10.1016/j.addbeh.2020.106778>
- Williams, G. C., Cole, A. G., de Groh, M., Jiang, Y., & Leatherdale, S. T. (2022). More support needed: Evaluating the impact of school e-cigarette prevention and cessation programs on e-cigarette initiation among a sample of Canadian secondary school students. *Preventive Medicine*, 155, 106924. <https://doi.org/10.1016/j.ypmed.2021.106924>
- Xie, W., Kathuria, H., Galiatsatos, P., Blaha, M. J., Hamburg, N. M., Robertson, R. M., Bhatnagar, A., Benjamin, E. J., & Stokes, A. C. (2020). Association of electronic cigarette use with incident respiratory conditions among US adults from 2013 to 2018. *JAMA Network Open*, 3(11), e2020816. <https://doi.org/10.1001/jamanetworkopen.2020.20816>
- Xuan, Z., & Choi, J. N. (2021). Content analysis of the use of fear in the real cost youth e-cigarette prevention campaign. *Journal of Communication in Healthcare*, 14(3), 206–215. <https://doi.org/10.1080/17538068.2020.1860671>
- Xu, Z., Dam, L., & Park, S. (2023). Using virtual reality in e-cigarette and secondhand aerosol prevention messages: Implications for emotional campaign design. *Cyberpsychology, Behavior and Social Networking*, 26(4), 279–287. <https://doi.org/10.1089/cyber.2022.0231>
- Zhang, Q. C., Courtney-Long, E. A., Sinclair, L. B., Reese, S., Armour, B. S., & Shapira, S. K. (2022). State-specific prevalence of current e-cigarette use by disability status and disability type—United States, BRFSS 2016–2018. *Disability and Health Journal*, 15(1), 101182. <https://doi.org/10.1016/j.dhjo.2021.101182>
- Zhao, X., Cai, X., & Malterud, A. (2023). Framing effects in youth e-cigarette use prevention: Individual text messages versus simulated text exchanges. *Health Education & Behavior*, 50(5), 683–692. <https://doi.org/10.1177/10901981221148965>